MODULE *Axioms*

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\begin{array}{l}
IntAxiom(t) \triangleq \\
\quad \forall i \in OpsOf(t) : \\
\qquad op\_seq[t][i].type = \text{"R"} \Rightarrow \\
\qquad\quad \textsc{let } prev\_ops \triangleq \\
\qquad\qquad\quad \{j \in 1 \,..\, (i-1) : op\_seq[t][j].key = op\_seq[t][i].key\} \\
\qquad\quad \textsc{in } \quad prev\_ops \neq \{\} \Rightarrow \\
\qquad\qquad\qquad \textsc{let } max\_j \triangleq \textsc{choose } j \in prev\_ops : \forall k \in prev\_ops : j \geq k \\
\qquad\qquad\qquad \textsc{in } \quad op\_seq[t][i].val = op\_seq[t][max\_j].val
\end{array}
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\begin{array}{l}
ExtAxiom(t, VIS, AR) \triangleq \\
\quad \forall x \in Keys, \\
\qquad v \in Values : \\
\qquad T\_reads\_ext(t, x, v) \Rightarrow \\
\qquad\quad \textsc{let } vis\_writes \triangleq \\
\qquad\qquad\quad \{tp \in CommittedTxns : \\
\qquad\qquad\qquad \langle tp, t\rangle \in VIS \wedge LastWriteIndex(tp, x) > 0 \\
\qquad\qquad\quad \} \\
\qquad\quad \textsc{in } \quad T\_writes(MaxAR(vis\_writes, AR), x, v)
\end{array}
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\begin{array}{l}
TransVisAxiom(t, VIS) \triangleq \\
\quad \forall tp, s \in CommittedTxns : \\
\qquad (\langle tp, s\rangle \in VIS \wedge \langle s, t\rangle \in VIS) \Rightarrow \langle tp, t\rangle \in VIS \\
PrefixAxiom(t, VIS, AR) \triangleq \\
\quad \forall tp, s \in CommittedTxns : \\
\qquad (\langle tp, s\rangle \in AR \wedge \langle s, t\rangle \in VIS) \Rightarrow \langle tp, t\rangle \in VIS \\
NoConflictAxiom(t, VIS, AR) \triangleq \\
\quad \forall tp \in CommittedTxns : \\
\qquad (WriteConflict(tp, t) \wedge \langle tp, t\rangle \in AR) \Rightarrow \langle tp, t\rangle \in VIS \\
TotalVisAxiom(t, VIS, AR) \triangleq \\
\quad \forall tp \in CommittedTxns : \langle tp, t\rangle \in AR \Rightarrow \langle tp, t\rangle \in VIS
\end{array}
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